

# Computer System

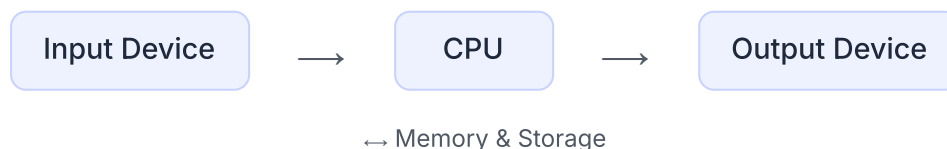
Information Technology · Class XI

## 1.1 Introduction to Computer System

A computer is an electronic device that can be programmed to accept data (**input**), process it and generate result (**output**). A computer along with additional hardware and software together is called a **computer system**.

A computer system primarily comprises of a central processing unit, memory, input/output devices, and storage devices. All these components function together as a single unit to deliver the desired output. A computer system comes in various forms and sizes — from a high-end server to a personal desktop, laptop, tablet, or smartphone.

Figure 1.1 — Components of a Computer System



Block diagram showing flow of data and signals between components.

### 1.1.1 Central Processing Unit (CPU)

It is the electronic circuitry of a computer that carries out the actual processing and is usually referred to as the **brain of the computer**. It is also commonly called **processor**. Physically, a CPU can be placed on one or more microchips called **integrated circuits (IC)**.

The CPU fetches the program and data from the memory, performs arithmetic and logical operations, and stores the result back to memory. While processing, the CPU stores data and instructions in its local memory called **registers**.

The CPU has two main components — **Arithmetic Logic Unit (ALU)** and **Control Unit (CU)**. ALU performs arithmetic and logic operations. CU controls sequential instruction execution and guides data flow.

### 1.1.2 Input Devices

The devices through which control signals are sent to a computer are termed as **input devices**. They convert input data into a digital form acceptable by the computer system. Examples include keyboard, mouse, scanner, touch screen, etc. Specially designed braille keyboards are also available for the visually impaired.

### 1.1.3 Output Devices

The device that receives data from a computer system for display or physical production is called **output device**. It converts digital information into human-understandable form. Examples: monitor, projector, headphone, speaker, printer, etc. A braille display monitor helps visually challenged individuals understand textual output.

Three types of commonly used printers are inkjet, laserjet and dot matrix. **3D printers** are used to build physical replicas of digital 3D designs in manufacturing and medical fields.

## 1.2 Evolution of Computer

From the simple calculator to a modern-day powerful data processor, computing devices have evolved in a relatively short span of time.

Figure 1.4 — Von Neumann Architecture



Control Unit guides the flow

ENIAC was the first binary programmable computer based on Von Neumann architecture.

In the 1970s, **Large Scale Integration (LSI)** allowed integration of a complete CPU on a single chip, called a **microprocessor**. Moore's Law predicted the exponential growth of transistors on a chip. In the 1980s, **Very Large Scale Integration (VLSI)** integrated around 3 million components. Further advancement led to **Super Large Scale Integration (SLSI)** with  $\approx 10^6$  components on a single IC.

IBM introduced its first PC in 1981, Apple introduced Macintosh in 1984. The popularity surged with **Graphical User Interface (GUI)** based operating systems. In the 1990s, the **World Wide Web (WWW)** accelerated mass usage. Today, we have smartphones, tablets, wearables, and IoT devices leveraging artificial intelligence.

## 1.3 Computer Memory

A computer system needs memory to store data and instructions for processing. **Primary memory** (main memory) and **secondary memory** (storage) serve different purposes.

### 1.3.1 Units of Memory

A computer uses binary numbers to store data. The binary digits **0** and **1** are called **bits**. A 4-bit word is a **Nibble**, and an 8-bit word is a **Byte**.

| Unit                 | Description       | Unit                  | Description    |
|----------------------|-------------------|-----------------------|----------------|
| <b>KB</b> (Kilobyte) | 1 KB = 1024 Bytes | <b>PB</b> (Petabyte)  | 1 PB = 1024 TB |
| <b>MB</b> (Megabyte) | 1 MB = 1024 KB    | <b>EB</b> (Exabyte)   | 1 EB = 1024 PB |
| <b>GB</b> (Gigabyte) | 1 GB = 1024 MB    | <b>ZB</b> (Zettabyte) | 1 ZB = 1024 EB |
| <b>TB</b> (Terabyte) | 1 TB = 1024 GB    | <b>YB</b> (Yottabyte) | 1 YB = 1024 ZB |

### 1.3.2 Types of Memory

#### (A) Primary Memory

**RAM (Random Access Memory)** is volatile — data is lost when power is turned off. It is used to store data temporarily while the computer is working. **ROM (Read Only Memory)** is non-volatile — contents are retained even when power is off. It stores the startup program (boot loader) that loads the OS into RAM.

#### (B) Cache Memory

A very high-speed memory placed between the CPU and primary memory. It stores copies of frequently accessed data, reducing the average time required to access data from primary memory.

#### (C) Secondary Memory

Non-volatile, larger storage capacity than primary memory, slower and cheaper. It cannot be accessed directly by the CPU; contents must first be brought into main memory. Examples: Hard Disk Drive (HDD), CD/DVD, Memory Card, Solid-State Drive (SSD), flash/pen drives.

### 1.3.3 Data Capturing, Storage, and Retrieval

**Data Capturing:** Gathering data from different sources in digital form (keyboard, barcode readers, remote sensors, social media comments, etc.).

**Data Storage:** Storing captured data for later processing. Large organisations use **data servers**.

**Data Retrieval:** Fetching data from storage devices for processing. Minimising data access time is crucial.

### 1.3.4 Data Deletion and Recovery

When data is deleted, its address entry is marked as free without actually removing the data. **Data recovery** is possible if the space marked as deleted has not been overwritten. Security concerns include unauthorised deletion and unwanted recovery. Use proper tools to **shred** data before disposing of old storage devices.

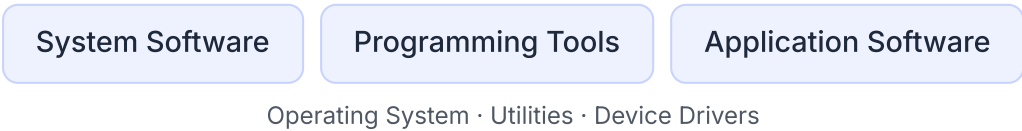
## 1.4 Software

**Hardware** refers to the physical components of a computer system which can be seen and touched. **Software** is a set of instructions and data that makes hardware functional to complete the desired task.

### 1.4.1 Need of Software

Software makes computer hardware useful and operational. It acts as an interface between human users and the hardware. Software can be broadly classified into three categories: **System Software**, **Programming Tools**, and **Application Software**.

Figure 1.9 — Categorisation of Software



## 1.4.2 System Software

Provides the basic functionality to operate a computer by interacting directly with hardware. Examples: **Operating Systems** (Windows, Linux, macOS, Ubuntu, Fedora, Android, iOS), **System Utilities** (disk defragmentation, formatting, anti-virus, disk cleaner), and **Device Drivers** (interface between OS and specific hardware devices).

## 1.4.3 Application Software

Works on top of system software to cater to specific end-user requirements. Two broad categories:

**General Purpose Software:** Ready-made software for generic applications. Examples: LibreOffice Calc, Adobe Photoshop, GIMP, Mozilla Firefox, iTunes.

**Customised Application Software:** Tailor-made for specific needs (e.g., banking software, hospital management systems).

## 1.4.4 Proprietary or Free and Open Source Software

**Free and Open Source Software (FOSS):** Source code is freely available for use and improvement. Examples: Ubuntu, Python, LibreOffice, Mozilla Firefox.

**Freeware:** Freely available for use but source code may not be available. Examples: Skype, Adobe Reader.

**Proprietary Software:** Must be purchased from the vendor who holds the copyright. Examples: Microsoft Windows, Tally, QuickHeal.

## Summary

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A computing device processes input data as per given instructions to generate desired output.

Computer system has four physical components: CPU, Primary memory, Input device, Output device.

Two types of primary memory: RAM (volatile) and ROM (non-volatile).

Software is a set of instructions categorised as system software, programming tools, and application software.

Operating system is an interface between the user and the computer, supervising hardware and software.

## Exercise

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Name the software required to make a computer functional. Write down its two primary functions.

What is the need of RAM? How does it differ from ROM?

What is the need for secondary memory?

Draw the block diagram of a computer system. Briefly write about the functionality of each component.

Differentiate between proprietary software and freeware software. Name two software of each type.

Mention any browsers used for browsing the internet.

Name the input/output device used to do the following:

- a) To output audio   b) To enter textual data   c) To make hard copy of a text file
- d) To display the data/information   e) To enter audio-based command
- f) To build 3D models   g) To assist a visually impaired individual in entering data

Identify the category (system, application, programming tool) of the following software:

- a) Compiler   b) Assembler   c) Ubuntu   d) Text editor

Convert the following into bytes:

- a) 2 MB   b) 3.7 GB   c) 1.2 TB

What is the security threats involved when we throw away electronic gadgets that are non-functional?

Write down the type of memory needed to do the following:

- a) To store data permanently   b) To execute the program
- c) To store the instructions which can not be overwritten.